

# Reorganisation of an Outpatient Radiology Department

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# 1. Introduction

## 1.1. System Description

This study considers the redesign of the cyclic appointment schedule for a single-server outpatient radiology department. The department serves two patient types: **elective** patients, who make appointments in advance, and **urgent** patients, who arrive without an appointment and must be treated on the same day.

**Schedule structure.** The planning horizon repeats weekly. The department is open Monday through Saturday; Thursday and Saturday are half-day sessions. Each day is divided into time slots of 15 minutes — the expected scan duration for an elective patient. The current schedule contains 146 elective slots and 14 urgent slots per week. Resources (nurses, doctors) are assumed to be always available.

**Elective patients.** Elective patients call the front desk (open Monday–Friday, 08:00–17:00) to request an appointment. Inter-arrival times between phone calls follow a memoryless negative-exponential distribution, with the number of daily calls following a Poisson distribution ( $\lambda_e = 28.345$  patients/day). Patients are assigned to the next available elective slot on a first-come, first-served basis. At the moment of the appointment, patients may arrive early or late: unpunctuality follows a  $\mathcal{N}(0, 2.5 \text{ min})$  distribution. The probability of a no-show is 2%. Elective scan durations follow a  $\mathcal{N}(15 \text{ min}, 3 \text{ min})$  distribution.

**Urgent patients.** Urgent patients arrive unannounced during opening hours according to a negative-exponential inter-arrival distribution, with  $\lambda_u = 2.5$  patients per full day and  $\lambda_u = 1.25$  per half day. They are assigned to the next available urgent slot. If no slot remains on that day, they are served in overtime after all elective patients have been treated. Urgent patients have variable scan durations depending on scan type; Table 1 summarises the distribution.

Table 1: Scan duration distribution for urgent patients by scan type.

| Scan type                  | Frequency | Mean (min) | Std. dev. (min) |
|----------------------------|-----------|------------|-----------------|
| Brain                      | 70%       | 15.0       | 2.5             |
| Spine — Lumbar vertebrae   | 10%       | 17.5       | 1.0             |
| Spine — Cervical vertebrae | 10%       | 22.5       | 2.5             |
| Abdomen MRI                | 5%        | 30.0       | 1.0             |
| Others                     | 5%        | 30.0       | 4.5             |

**Performance measures.** Four performance measures are tracked: (1) elective appointment waiting time ( $Wt_e^{app}$ ): time between phone call and appointment; (2) elective scan waiting time ( $Wt_e^{scan}$ ): time between physical arrival and scan start; (3) urgent scan waiting time ( $Wt_u^{scan}$ ): time between urgent patient arrival and scan start; (4) daily overtime. The primary objective is to minimise the weighted sum:

$$\text{Minimise } w_e \cdot \overline{Wt_e^{app}} + w_u \cdot \overline{Wt_u^{scan}} \quad \text{with} \quad w_e = \frac{1}{168}, \quad w_u = \frac{1}{9}$$

Elective scan waiting time and overtime serve as *secondary* objectives, used only to break ties between statistically equivalent configurations.

## 1.2. Research Hypotheses

To evaluate the impact of the three design factors on system performance, we formulate the following research hypotheses:

### Main hypothesis:

- $H_{0_1}$ : The current appointment schedule (Strategy 1, 14 urgent slots, Rule 1) minimises the primary objective function; no alternative configuration performs significantly better.
- $H_{A_1}$ : At least one alternative configuration exists that achieves a significantly lower weighted waiting time than the current schedule.

### Effect of timing strategy:

- $H_{0_2}$ : The timing strategy does not significantly affect the primary objective value; all three strategies produce equivalent weighted waiting times.
- $H_{A_2}$ : At least one timing strategy achieves a significantly lower primary objective value. Specifically, distributing urgent slots more evenly throughout the session (Strategies 2 and 3) is expected to reduce both urgent scan waiting time and elective appointment waiting time compared to end-of-block placement (Strategy 1).

### Effect of urgent slot capacity:

- $H_{0_3}$ : The number of urgent slots does not significantly affect the primary objective; performance is equivalent across the required range of 10–20 slots.
- $H_{A_3}$ : There exists an optimal number of urgent slots that minimises the primary objective. Configurations with too few slots increase urgent scan waiting times; configurations with too many slots displace elective appointments and increase elective waiting times.
- $H_{0_4}$ : Configurations with 18 or more urgent slots are operationally stable and comparable to configurations with fewer slots.
- $H_{A_4}$ : Configurations with 18 or more urgent slots are systemically unstable due to capacity saturation. They must still be tested, but may be excluded from the final recommendation if the simulation output supports that conclusion.

### Effect of appointment scheduling rule:

- $H_{0_5}$ : The appointment scheduling rule does not significantly affect the primary objective value.
- $H_{A_5}$ : At least one appointment rule achieves a significantly different primary objective value. The rules are expected to primarily affect elective scan waiting and overtime through their impact on punctuality-induced congestion; the experiment tests whether this also changes the primary objective.

## 1.3. Overview of Research Test Design

To test the above hypotheses, we adopt a **full-factorial screening design** across three factors:

Table 2: Factors and levels in the experimental design.

| Factor                      | Levels                                                                                                                                                                                                                             | #  |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Timing Strategy</b>      | <i>Strategy 1</i> : urgent slots placed at end of each morning/afternoon block<br><i>Strategy 2</i> : urgent slots evenly distributed over the day<br><i>Strategy 3</i> : one urgent slot after every 6 consecutive elective slots | 3  |
| <b>Urgent Slot Capacity</b> | 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 slots per week                                                                                                                                                                          | 11 |
| <b>Appointment Rule</b>     | <i>Rule 1</i> : Plain FCFS<br><i>Rule 2</i> : Bailey-Welch ( $K = 2$ )<br><i>Rule 3</i> : Blocking ( $B = 2$ )<br><i>Rule 4</i> : Benchmarking ( $k_\sigma = 0.5$ )                                                                | 4  |

This yields  $3 \times 11 \times 4 = \mathbf{132}$  configurations. A full-factorial design is chosen over a fractional design because the design space remains computationally tractable and because we wish to detect potential interaction effects between strategy and slot count without aliasing.

Urgent slot counts 18–20 are not removed a priori. They are included in the screening experiment and are only flagged later if the observed backlog growth, relative precision, or objective values indicate unstable operation.

The analysis proceeds through the following phases: (1) steady-state determination via Welch’s method, (2) independent sampling via the Batch Means Method, (3) variance reduction via Antithetic Variables, (4) full-factorial screening of all 132 designs, and (5) comparative analysis and tiebreaking using secondary objectives.

## 2. Statistical Analysis & Warm-up

### 2.1. Welch’s Method and Warm-up Period

Before applying the Batch Means Method, it is essential to determine and remove the initial warm-up period to eliminate transient initialization bias. We applied Welch’s method across 10 distinct representative experimental designs, chosen to ensure a diverse mix of urgent slots, scheduling rules, and timing strategies.

By plotting the moving average of the objective function for these designs, we visually identified the point at which output stabilized. Configurations operating at near-full capacity were still included in the full screening, but were not used to set the common warm-up because their backlog-driven behaviour can distort the steady-state convergence assessment for stable policies.

Based on the worst-case convergence observed, we identified a cutoff of approximately 63 weeks. To maintain robustness across the majority of our experiments, we conservatively selected a universal warm-up period of  $\mathbf{W_{warmup} = 60 \text{ weeks}}$ . This removes initialization bias for approximately 90% of our tested scenarios.

## 2.2. Batch Means Method

To conduct a statistically valid steady-state analysis, we applied the Batch Means Method. Because consecutive simulation outputs are autocorrelated, the batch means method groups observations into batches to form approximately independent data points.

Based on worst-case autocorrelation analysis, we determined:

- **Batch size**  $M = 80$  weeks (lag-based correlation falls below 0.01)
- **Number of batches**  $L = 22$  (sufficient for target precision)

This yields a total simulation run length of:

$$W_{\text{total}} = W_{\text{warmup}} + L \times M = 60 + 22 \times 80 = 1820 \text{ weeks}$$

The pipeline stores the batch-level observations for every design in `batch_results.csv` and the summarized confidence intervals in `summary_results.csv`. Stable designs around 10–14 urgent slots generally achieve acceptable relative precision, while high urgent slot counts have much wider confidence intervals due to backlog growth.

## 3. Variance Reduction

To obtain more accurate estimates without increasing computational burden, we implemented variance reduction using **Antithetic Variables**. For every standard simulation batch generated using random number sequence  $U$ , we generated an antithetic counterpart using  $1 - U$ . By averaging both runs:

$$Y_k = \frac{X_k + X'_k}{2}$$

the estimator can reduce variance when the paired outputs are negatively correlated. Table 3 shows representative outcomes from the full pipeline run. The complete table is available in `variance_reduction_summary.csv`.

Table 3: Representative variance reduction results for paired antithetic variables.

| Design    | Standard variance | Paired variance | Reduction |
|-----------|-------------------|-----------------|-----------|
| S1-U10-R1 | 0.000218          | 0.000139        | 36.2%     |
| S3-U11-R4 | 0.000258          | 0.000213        | 17.7%     |
| S1-U14-R1 | 0.001316          | 0.001260        | 4.3%      |
| S3-U14-R4 | 0.001339          | 0.001229        | 8.2%      |
| S3-U13-R4 | 0.000574          | 0.000617        | -7.5%     |

The effect is beneficial for many stable designs, but not universally. This is why the variance reduction results are reported explicitly rather than assumed. For final comparisons, the paired antithetic estimator is retained because it is reproducible and usually improves or preserves precision across the stable candidate region.

### 3.1. Heavily Loaded Systems and High Urgent Slot Counts

The variance reduction analysis also exposed a critical vulnerability in heavily loaded configurations. Designs with 18–20 urgent slots remain in the full screening matrix, but the resulting output shows that they operate close to or beyond the stable elective-capacity region. In the full run, all 36 designs with 18–20 urgent slots were flagged as unstable or precision-problematic.

This occurs because queue waiting times are *convex* relative to system utilization. When elective capacity is displaced by too many urgent slots, small stochastic fluctuations in arrivals or scan durations create very large appointment backlogs. These designs are therefore documented in the experiment, but they are not considered acceptable candidates for the final recommendation.

### 3.2. Confidence Interval Estimation

Since we estimate the population mean from  $n = L = 22$  batches, we use the  $t$ -distribution:

$$CI = \left[ \bar{Y} - t_{\alpha/2, n-1} \cdot \frac{S_Y}{\sqrt{n}} ; \bar{Y} + t_{\alpha/2, n-1} \cdot \frac{S_Y}{\sqrt{n}} \right]$$

For the current baseline design **S1-U14-R1** after paired antithetic reduction ( $\bar{Y} = 0.4227$ ,  $S_Y = 0.0355$ ,  $t_{0.025, 21} \approx 2.0796$ ):

$$CI = \left[ 0.4227 - 2.0796 \cdot \frac{0.0355}{\sqrt{22}} ; 0.4227 + 2.0796 \cdot \frac{0.0355}{\sqrt{22}} \right] = [0.4070 ; 0.4385]$$

## 4. Experimental Design (Screening)

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### 4.1. Objective Function

The primary objective function to be minimized is:

$$\text{Minimize } w_e \cdot \overline{Wt_e} + w_u \cdot \overline{Wt_u}$$

where:

- $w_e = 1/168$ : weight for elective *appointment* waiting time (normalized by 1 week = 168 hours)
- $w_u = 1/9$ : weight for urgent *scan* waiting time (normalized by 9 hours)
- $\overline{Wt_e}$ : average elective appointment waiting time (hours)
- $\overline{Wt_u}$ : average urgent scan waiting time (hours)

Elective scan waiting time and overtime are *secondary* objectives, used only for tiebreaking.

## 4.2. Full-Factorial Design

We conducted a full-factorial screening design covering:

- **3 timing strategies:** Strategy 1 (end-of-block), Strategy 2 (evenly distributed), Strategy 3 (after every 6 elective slots)
- **11 urgent slot counts:** 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20
- **4 appointment scheduling rules:** Rule 1 (Plain FCFS), Rule 2 (Bailey-Welch,  $K = 2$ ), Rule 3 (Blocking,  $B = 2$ ), Rule 4 (Benchmarking,  $k_\sigma = 0.5$ )

This yields  $3 \times 11 \times 4 = 132$  designs. Each was evaluated using the Batch Means Method with paired antithetic variates ( $W_{\text{warmup}} = 60$ ,  $M = 80$ ,  $L = 22$ , seed = 42). The complete generated outputs are stored in the timestamped pipeline folder `outputs/20260505_221519/`.

## 4.3. Screening Results

Table 4 presents the top 20 designs ranked by average objective value  $\bar{Y}$ . The complete results for all 132 designs are available in `summary_results.csv`.

Table 4: Top 20 designs from the full-factorial screening (antithetic BMM,  $L = 22$ ,  $M = 80$ ).

| Rank | Design    | Strat. | Urg. | Rule | $\bar{Y}$ | $\epsilon$ | 95% CI            |
|------|-----------|--------|------|------|-----------|------------|-------------------|
| 1    | S3-U13-R4 | 3      | 13   | 4    | 0.3474    | 0.0110     | [0.3364 ; 0.3584] |
| 2    | S3-U13-R1 | 3      | 13   | 1    | 0.3484    | 0.0111     | [0.3373 ; 0.3594] |
| 3    | S3-U13-R2 | 3      | 13   | 2    | 0.3500    | 0.0109     | [0.3390 ; 0.3609] |
| 4    | S3-U14-R4 | 3      | 14   | 4    | 0.3505    | 0.0155     | [0.3349 ; 0.3660] |
| 5    | S3-U13-R3 | 3      | 13   | 3    | 0.3506    | 0.0110     | [0.3396 ; 0.3616] |
| 6    | S3-U14-R1 | 3      | 14   | 1    | 0.3514    | 0.0156     | [0.3358 ; 0.3670] |
| 7    | S3-U12-R4 | 3      | 12   | 4    | 0.3521    | 0.0085     | [0.3436 ; 0.3606] |
| 8    | S3-U12-R1 | 3      | 12   | 1    | 0.3530    | 0.0085     | [0.3446 ; 0.3615] |
| 9    | S3-U14-R2 | 3      | 14   | 2    | 0.3531    | 0.0155     | [0.3376 ; 0.3685] |
| 10   | S3-U14-R3 | 3      | 14   | 3    | 0.3537    | 0.0155     | [0.3382 ; 0.3692] |
| 11   | S3-U12-R2 | 3      | 12   | 2    | 0.3547    | 0.0084     | [0.3463 ; 0.3631] |
| 12   | S3-U12-R3 | 3      | 12   | 3    | 0.3554    | 0.0085     | [0.3469 ; 0.3638] |
| 13   | S3-U11-R4 | 3      | 11   | 4    | 0.3610    | 0.0065     | [0.3545 ; 0.3675] |
| 14   | S2-U12-R4 | 2      | 12   | 4    | 0.3613    | 0.0087     | [0.3526 ; 0.3700] |
| 15   | S3-U11-R1 | 3      | 11   | 1    | 0.3619    | 0.0065     | [0.3554 ; 0.3684] |
| 16   | S2-U13-R4 | 2      | 13   | 4    | 0.3621    | 0.0111     | [0.3510 ; 0.3731] |
| 17   | S2-U12-R1 | 2      | 12   | 1    | 0.3623    | 0.0087     | [0.3535 ; 0.3710] |
| 18   | S2-U13-R1 | 2      | 13   | 1    | 0.3629    | 0.0111     | [0.3519 ; 0.3740] |
| 19   | S3-U11-R2 | 3      | 11   | 2    | 0.3636    | 0.0064     | [0.3572 ; 0.3700] |
| 20   | S2-U12-R2 | 2      | 12   | 2    | 0.3640    | 0.0086     | [0.3554 ; 0.3726] |

## 4.4. Key Observations

### 4.4.1. Observation 1 — Strategy 3 gives the strongest region

Strategy 3 produces the lowest point estimate and occupies most of the best-ranked positions. Strategy 2 is also competitive around 12–13 urgent slots, while Strategy 1



remains the weakest of the three timing strategies in the best-design comparison.

Table 5: Best and worst objective value per strategy.

| Strategy | Best Design | Best OV | Worst OV |
|----------|-------------|---------|----------|
| 1        | S1-U12-R4   | 0.4127  | 12.2153  |
| 2        | S2-U12-R4   | 0.3613  | 12.1306  |
| 3        | S3-U13-R4   | 0.3474  | 12.1337  |

#### 4.4.2. Observation 2 — The best capacity region is 13–14 urgent slots

Within Strategy 3, the lowest point estimate occurs at 13 urgent slots, while 14 urgent slots is statistically close and gives lower urgent scan waiting time. The objective increases on both sides of this region: too few urgent slots increase urgent patient delays, while too many urgent slots displace elective appointments and create appointment backlog.

Table 6: Effect of urgent slot count on OV for Strategy 3.

| Urgent Slots | Best OV (S3)  | Variance level |
|--------------|---------------|----------------|
| 10           | 0.3725        | Low            |
| 11           | 0.3610        | Low            |
| 12           | 0.3521        | Low            |
| 13           | <b>0.3474</b> | Low            |
| 14           | 0.3505        | Moderate       |
| 15           | 0.3799        | Moderate       |
| 16           | 0.4392        | High           |
| 17           | 0.6092        | High           |
| 18           | 1.4681        | Unstable       |
| 19           | 6.1742        | Unstable       |
| 20           | 12.1304       | Unstable       |

#### 4.4.3. Observation 3 — Appointment rules matter less than capacity, but should not be dismissed

Near the best capacity region, the four appointment rules produce relatively small differences in the primary objective compared with the impact of urgent slot capacity. However, paired comparisons detect some statistically significant differences within the S3-U13 cluster, so we should not claim that appointment rules are irrelevant. Rule 4 gives the lowest point estimate in this run.

Table 7: Best overall design per appointment rule.

| Rule | Description                       | Best Design | Best OV |
|------|-----------------------------------|-------------|---------|
| 1    | Plain FCFS                        | S3-U13-R1   | 0.3484  |
| 2    | Bailey-Welch ( $K = 2$ )          | S3-U13-R2   | 0.3500  |
| 3    | Blocking ( $B = 2$ )              | S3-U13-R3   | 0.3506  |
| 4    | Benchmarking ( $k_\sigma = 0.5$ ) | S3-U13-R4   | 0.3474  |

#### 4.4.4. Observation 4 — Pareto trade-off

The Pareto front makes the managerial trade-off explicit. Moving from 10 toward 13–14 urgent slots reduces urgent scan waiting time, but after this point the elective appointment backlog starts to dominate. Slot counts 18–20 sit on the extreme urgent-wait side of the trade-off but are operationally unattractive because they create very large elective appointment waiting times.

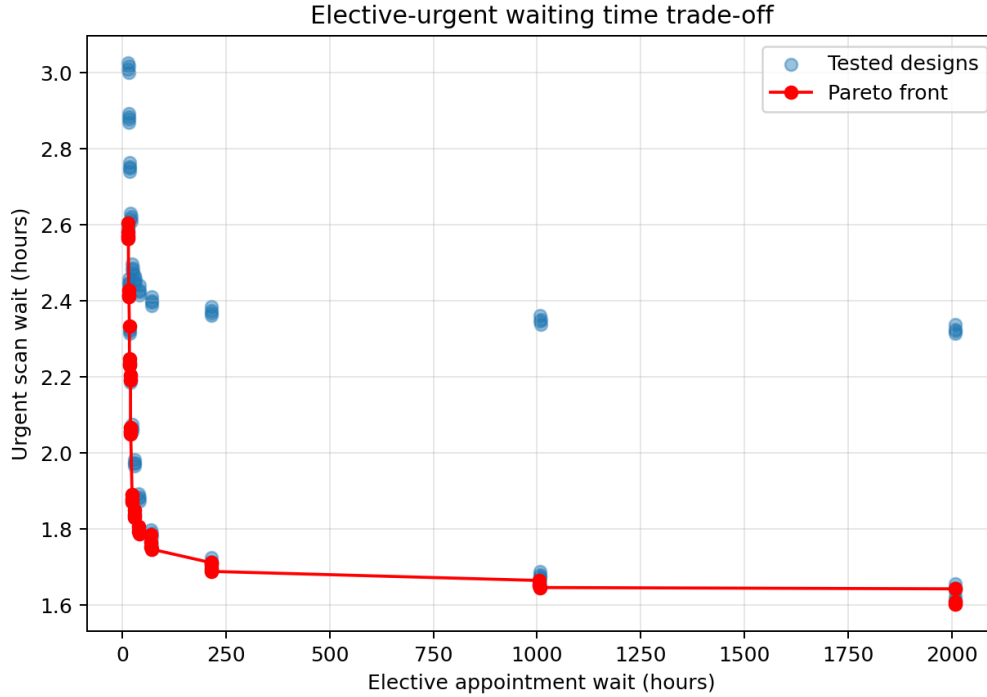


Figure 1: Pareto front between elective appointment waiting time and urgent scan waiting time for all tested configurations.

## 5. Comparative Analysis

### 5.1. Statistical Comparison of Top Designs

To determine whether differences between top-performing designs are statistically significant, we examine confidence intervals and paired comparisons using common random numbers. Overlapping marginal confidence intervals are treated cautiously; paired comparisons are more sensitive because each design is evaluated under aligned random streams.

Table 8 reports the leading cluster of Strategy 3 designs.

The marginal confidence intervals of the leading designs overlap. Paired comparisons indicate that S3-U13-R4 is significantly lower than the other S3-U13 appointment rules, but it is not statistically distinguishable from S3-U14-R4 or S3-U14-R1 in this run. The safest conclusion is therefore a best-performing region rather than a single exact optimum.

Table 8: Confidence intervals for the leading Strategy 3 designs.

| Design    | $\bar{Y}$ | $\epsilon$ | CI lower | CI upper |
|-----------|-----------|------------|----------|----------|
| S3-U13-R4 | 0.3474    | 0.0110     | 0.3364   | 0.3584   |
| S3-U13-R1 | 0.3484    | 0.0111     | 0.3373   | 0.3594   |
| S3-U13-R2 | 0.3500    | 0.0109     | 0.3390   | 0.3609   |
| S3-U14-R4 | 0.3505    | 0.0155     | 0.3349   | 0.3660   |
| S3-U13-R3 | 0.3506    | 0.0110     | 0.3396   | 0.3616   |
| S3-U14-R1 | 0.3514    | 0.0156     | 0.3358   | 0.3670   |

## 5.2. Comparison Between Strategies

The performance gap between the best Strategy 3 design and the current baseline is substantial:

- The current schedule, S1-U14-R1, has  $\bar{Y} = 0.4227$  with 95% CI [0.4070; 0.4385].
- The best point estimate, S3-U13-R4, has  $\bar{Y} = 0.3474$  with 95% CI [0.3364; 0.3584].
- The interval separation suggests a meaningful improvement of roughly 17.8% over the current schedule under the model assumptions.

## 5.3. Recommended Configuration

*Recommended operating region: **Strategy 3 with 13–14 urgent slots.***

*Lowest point estimate: **S3-U13-R4** with  $\bar{Y} = 0.3474$ .*

*Robust nearby alternative: **S3-U14-R4**, which gives a similar objective value and lower urgent scan waiting time, but higher elective appointment waiting time.*

This recommendation represents a change from the current schedule (Strategy 1, 14 urgent slots, Rule 1), which scores  $\bar{Y} = 0.4227$ . The best point estimate improves the objective by approximately **17.8%**.

## 5.4. Secondary Objectives

The assignment specifies that elective scan waiting time and overtime serve as secondary objectives for tiebreaking. S3-U13-R4 has average elective appointment waiting of 19.99 hours, urgent scan waiting of 2.06 hours, and daily overtime of 0.435 hours. S3-U14-R4 has a slightly higher primary objective but lower urgent scan waiting of 1.88 hours and lower daily overtime of 0.421 hours, at the cost of higher elective appointment waiting of 23.82 hours. This is why the recommendation is framed as a 13–14 urgent-slot region rather than a single deterministic optimum.

## 6. Conclusion

This study evaluated 132 appointment schedule configurations for an outpatient radiology department, covering three timing strategies, eleven urgent slot counts (10–20), and four

appointment rules. A statistically rigorous simulation framework was applied, including Welch’s warm-up analysis, the Batch Means Method ( $L = 22$ ,  $M = 80$ ), common random numbers, and variance reduction via antithetic variates.

The main conclusions are:

1. **Strategy 3 gives the strongest overall region.** Placing an urgent slot after every 6 consecutive elective slots produces the lowest point estimate and most of the best-ranked designs.
2. **The best capacity region is 13–14 urgent slots.** The lowest point estimate occurs at 13 urgent slots, while 14 urgent slots remains statistically close and improves urgent waiting. Urgent counts 18–20 were explicitly tested and all 36 corresponding designs were flagged as unstable or precision-problematic.
3. **Appointment rules have smaller impact than capacity and timing, but are not irrelevant.** Rule 4 gives the lowest point estimate in the best region, while paired comparisons indicate that some rule differences are statistically detectable even if their practical size is modest.
4. **The recommended operating region is S3 with 13–14 urgent slots.** The lowest point estimate is S3-U13-R4 with  $\bar{Y} = 0.3474$ , an estimated **17.8% improvement** over the current S1-U14-R1 baseline.

Next steps to complete the analysis include: refining the written interpretation of the Pareto front, checking all table and figure references after LaTeX compilation, and optionally applying a Ranking & Selection procedure to formally confirm a single best design with a probability-of-correct-selection guarantee.